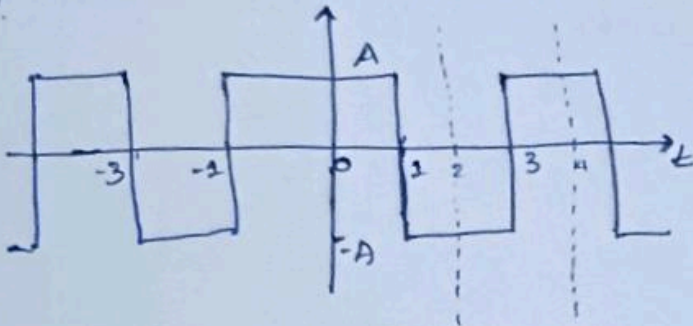


EE3203 : Power Electronics

TUTORIAL 1 : FOURIER ANALYSIS

1 (a)



[0, 4] is the period

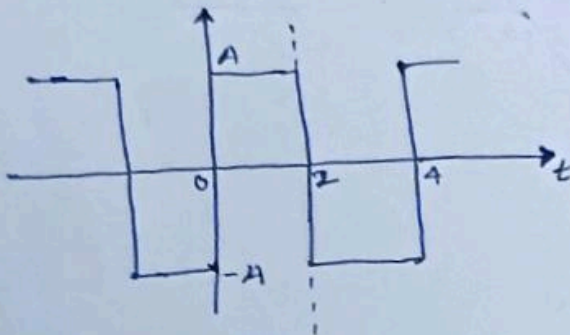
Symmetric about y-axis

 $f(-x) = f(x) \rightarrow$ even symmetry $f(x-2) = -f(x)$

cosine terms + odd harmonics

[half wave symmetry]

(b)

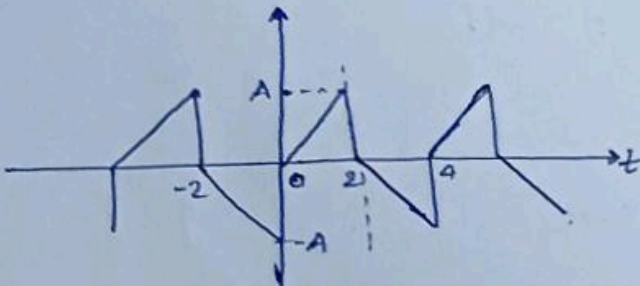


Symmetric about y-axis

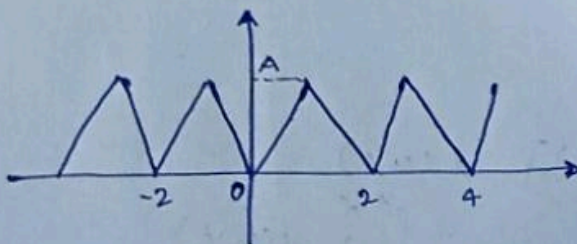
Odd symmetry and
Half wave symmetry

Sine terms + Odd harmonics

(c)

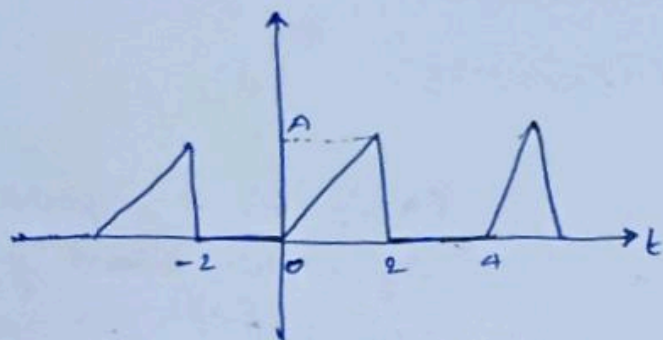
NOT symmetric about
x and y axes but
half wave symmetry exists

(d)



Only even symmetry [@ 4(b)]

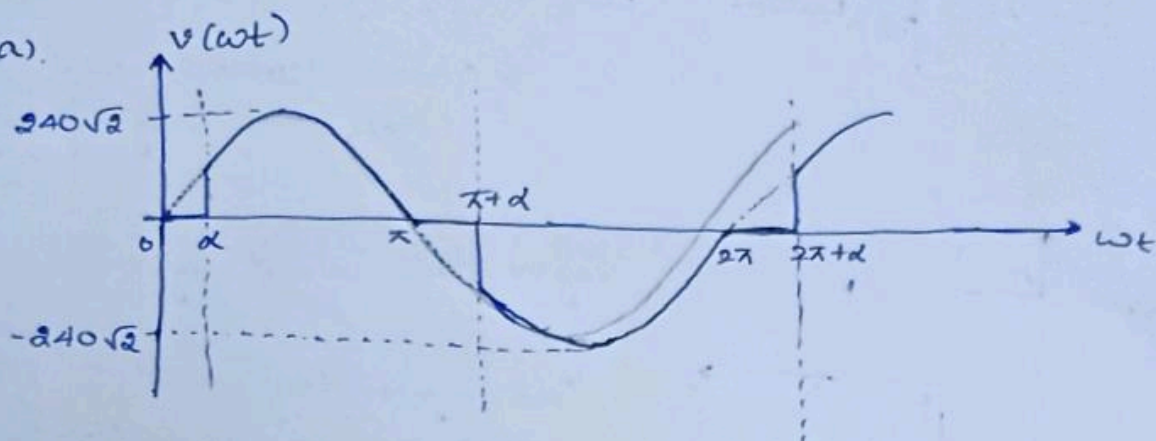
(e).



Neither odd nor even symmetry

It doesn't possess half wave symmetry

2. (a).



- Neither even nor odd symmetry
- Half wave symmetry
- Contains sine and cosine terms but only odd harmonics, and DC component is zero

$$\Rightarrow a_0 = 0, \begin{cases} a_n \neq 0 \\ b_n \neq 0 \end{cases} \text{ for } n \text{ is odd}$$

$$a_n = \frac{4}{T} \int_0^{T/2} f(\omega t) \cos n\omega t d(\omega t)$$

$$\Rightarrow a_n = \frac{4}{2\pi} \int_{\alpha}^{\pi+\alpha} V_m \sin \omega t \cos n\omega t d(\omega t)$$

$$a_n = \frac{2}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t \cos n\omega t d(\omega t)$$

$$a_n = \frac{2V_m}{2\pi} \int_{\alpha}^{\pi} [\sin(1+n)\omega t + \sin(1-n)\omega t] d(\omega t) + \int_{\pi}^{\pi+\alpha} 0 d(\omega t)$$

$$a_n = \frac{V_m}{\pi} \int_{\alpha}^{\pi} \sin \omega t d(\omega t)$$

$$\sin a \cos b$$

$$= \frac{1}{2} \left\{ \sin(a+b) + \sin(a-b) \right\}$$

$$\sin(-x) = -\sin x$$

$$a_n = \frac{2V_m}{2\pi} \int_{\alpha}^{\pi} [\sin(n+\frac{1}{2})\omega t - \sin(n-\frac{1}{2})\omega t] d(\omega t) + \frac{2V_m}{2\pi} \int_{\pi}^{\pi+\alpha} 0 d(\omega t)$$

$\left. \begin{array}{l} \sin a \cos b \\ = \frac{1}{2} \{ \sin(a+b) + \sin(a-b) \} \\ \sin(-x) = -\sin x \end{array} \right\}$

$$a_n = \frac{V_m}{\pi} \left[\frac{-\cos(n+1)\omega t}{(n+1)} + \frac{\cos(n-1)\omega t}{(n-1)} \right]_{\alpha}^{\pi}$$

$$= \frac{V_m}{\pi} \left[\frac{1}{n+1} [-(-1)^{n+1}] + \frac{1}{n-1} [(-1)^{n-1}] \right.$$

$$\left. + \frac{1}{n+1} \cos(n+1)\alpha - \frac{\cos(n-1)\alpha}{n-1} \right]$$

$$= \frac{V_m}{\pi} \left\{ \frac{\cos(n+1)\alpha}{n+1} - \frac{\cos(n-1)\alpha}{n-1} + \frac{(-1)^n (-2)}{n^2-1} \right\}$$

$$\Rightarrow a_n = \frac{240\sqrt{2}}{\pi} \left\{ \frac{\cos(n+1)\alpha}{(n+1)} - \frac{\cos(n-1)\alpha}{(n-1)} + \frac{2(-1)^{n+1}}{(n^2-1)} \right\}$$

$$b_n = \frac{4}{T} \int_0^{T/2} f(\omega t) \sin n\omega_0 t d(\omega t)$$

$$\Rightarrow b_n = \frac{4}{2\pi} \int_{\alpha}^{\pi+\alpha} V_m \sin \omega t \sin n\omega_0 t d(\omega t)$$

$$= \frac{V_m}{\pi} \int_{\alpha}^{\pi} 2 \sin \omega t \sin n\omega_0 t d(\omega t) + \int_{\pi}^{\pi+\alpha} \frac{V_m}{\pi} (0) d(\omega t)$$

$$= \frac{V_m}{\pi} \int_{\alpha}^{\pi} [\cos(n-1)\omega t - \cos(n+1)\omega t] d(\omega t)$$

$$= \frac{V_m}{\pi} \left[\frac{\sin(n-1)\omega t}{(n-1)} - \frac{\sin(n+1)\omega t}{(n+1)} \right]_{\alpha}^{\pi}$$

$$\left. \begin{array}{l} \sin a \sin b \\ = \frac{1}{2} \{ \cos(a+b) - \cos(a-b) \} \\ \Rightarrow 2 \sin a \sin b \\ = \{ \cos(a-b) - \cos(a+b) \} \end{array} \right\}$$

$$b_n = \frac{V_m}{\pi} \left\{ \frac{1}{(n+1)} [-(-1)] \right\}$$

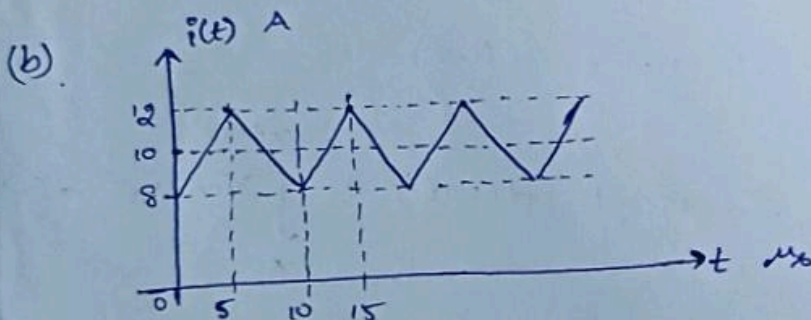
$$b_n = \frac{V_m}{\pi} \left\{ \frac{\sin(n+1)\alpha}{(n+1)} - \frac{\sin(n-1)\alpha}{(n-1)} \right\}$$

$$\Rightarrow b_n = \frac{240\sqrt{2}}{\pi} \left[\frac{\sin(n+1)\alpha}{(n+1)} - \frac{\sin(n-1)\alpha}{(n-1)} \right]$$

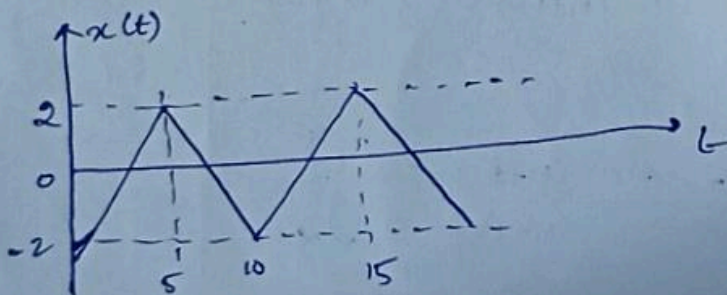
$$v(\omega t) = 0 + \sum_{n=1,3,5,\dots} \frac{240\sqrt{2}}{\pi} \left[\frac{\cos(n+1)\alpha}{(n+1)} - \frac{\cos(n-1)\alpha}{(n-1)} + \frac{2(-1)^{n+1}}{n^2-1} \right] \cos n\omega t$$

$$+ \sum_{n=1,3,5,\dots} \frac{240\sqrt{2}}{\pi} \left[\frac{\sin(n+1)\alpha}{(n+1)} - \frac{\sin(n-1)\alpha}{(n-1)} \right] \sin n\omega t$$

α	V RMS of Fundamental	V RMS of 3 rd harmonic
30°	233.8475 233.8475	38.1975 19.0975
45°	218.18 218.18	38.1975
60°	194.01 194.01	57.295



DC component,
 $a_0 = 10$



- Even symmetry
- Half wave symmetry
 $\Rightarrow b_n = 0$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{2 - (-2)}{5 \times 10^{-6}} = \frac{4}{5} \times 10^6 = m$$

$$y = mx + c \Rightarrow c = y - mx \quad \Big|_{(0, -2)}$$

$$\Rightarrow c = (-2) - \left(\frac{4}{5} \times 10^6\right)(0) = -2$$

$$\therefore x(t) = \left(\frac{4}{5} \times 10^6\right)t - 2$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos n\omega_0 t \, dt = \frac{4}{T} \int_0^{T/2} x(t) \cos n\omega_0 t \, dt$$

$$\Rightarrow a_n = \frac{4}{10\mu} \int_0^{5\mu} \left[\left(\frac{4}{5} \times 10^6\right)t - 2\right] \cos n\omega_0 t \, dt$$

$$\int_0^{5\mu} \left(\frac{4}{5} \times 10^6\right)t \cdot \cos n\omega_0 t \, dt$$

$$= \frac{4}{5} \times 10^6 \int_0^{5\mu} t \cdot \cos n\omega_0 t \, dt$$

$$= \frac{4}{5} \times 10^6 \left[\frac{t \sin n\omega_0 t}{n\omega_0} - \int \frac{\sin n\omega_0 t}{n\omega_0} \, dt \right]_0^{5\mu}$$

$$= \frac{4}{5} \times 10^6 \left[\frac{(5 \times 10^{-6}) \sin n \left(\frac{2\pi}{10\mu}\right) 5\mu}{n \left(\frac{2\pi}{10\mu}\right)} - 0 \right]$$

$$+ \left(\frac{1}{n\omega_0}\right)^2 \left[\cos n\omega_0 \left(\frac{2\pi}{10\mu}\right) 5\mu - 1 \right]$$

$$= \frac{4}{5} \times 10^6 \left[\frac{1}{n^2 \left(\frac{2\pi}{10\mu}\right)^2} (\cos n\pi - 1) \right]$$

$$= \frac{4 \times 10^6}{5n^2 \left(\frac{4\pi^2}{100}\right) \times 10^6 \times 10^6} [(-1)^n - 1]$$

$$a_n = \frac{100 \times 10^{-6}}{5n^2 \pi^2} [(-1)^n - 1]$$

$$\int_0^{5\mu} 2 \cos n\omega_0 t \, dt$$

$$= \frac{2}{n\omega_0} \left[\sin n\omega_0 t \right]_0^{5\mu}$$

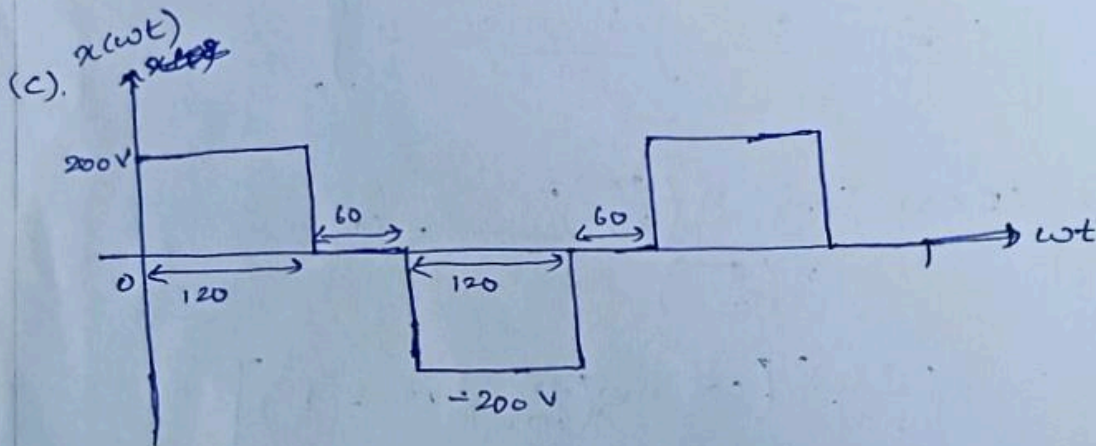
$$= \frac{2}{n\left(\frac{2\pi}{10\mu}\right)} \left[\sin n\left(\frac{2\pi}{10\mu}\right) 5\mu - 0 \right] = 0$$

$$\Rightarrow a_n = \frac{4}{10\mu} \left[\frac{100\mu}{5\pi^2 n^2} ((-1)^n - 1) - 0 \right]$$

$$\Rightarrow a_n = \frac{8}{\pi^2 n^2} [(-1)^n - 1] = \frac{-16}{\pi^2 n^2} \quad \text{for } n \text{ is odd}$$

$$\Rightarrow f(\omega t)$$

$$\Rightarrow i(t) = 10 + \sum_{n=1,3,5,\dots} \frac{-16}{n^2 \pi^2} \cos n\omega_0 t$$



$$f = 50 \text{ Hz}$$

DC component, $a_0 = 0$

$$a_n = \frac{2}{2\pi} \left[\int_0^{2\pi/3} V_m \cos nt \, dt - \int_{\pi}^{5\pi/3} V_m \cos nt \, dt \right] \quad \left| \begin{array}{l} a_n = \frac{2}{T} \int_0^T x(t) \cos n\omega_0 t \, dt \\ T = 2\pi; \omega_0 = 1 \end{array} \right.$$

$$= \frac{V_m}{n\pi} \left\{ \left[\sin \left(\frac{2\pi}{3} \right) n - 0 \right] - \left[\sin n \left(\frac{5\pi}{3} \right) - \sin n\pi \right] \right\}$$

$$= \frac{V_m}{n\pi} \left[\sin n \left(\frac{2\pi}{3} \right) - \sin n \left(\frac{5\pi}{3} \right) \right]$$

$$n=1, a_1 = \frac{V_m}{\pi} \left[\sin \frac{2\pi}{3} - \sin \frac{5\pi}{3} \right] = \frac{\sqrt{3}V_m}{\pi}$$

$$n=2, a_2 = \frac{V_m}{2\pi} \left[\sin \frac{4\pi}{3} - \sin \frac{10\pi}{3} \right] = 0$$

$$n=3, a_3 = \frac{V_m}{3\pi} \left[\sin 2\pi - \sin 5\pi \right] = 0$$

$$a_n = \frac{200}{n\pi} \left[\sin \frac{2\pi n}{3} - \sin \frac{5\pi n}{3} \right] ; n=1, 5, 7, 11, \dots$$

$$b_n = \frac{2}{2\pi} \left[\int_0^{2\pi/3} V_m \sin nt \, dt - \int_{\pi}^{5\pi/3} V_m \sin nt \, dt \right]$$

$$= \frac{V_m}{n\pi} \left[-\cos \frac{2\pi n}{3} + 1 + \cos \frac{5\pi n}{3} - \cos n\pi \right]$$

$$= \frac{200}{n\pi} \left[1 - (-1)^n + \cos \frac{5\pi n}{3} - \cos \frac{2\pi n}{3} \right]$$

$$n=1, b_1 = \frac{V_m}{\pi} \left[1+1 + \cos \frac{5\pi}{3} - \cos \frac{2\pi}{3} \right] = \frac{3V_m}{\pi}$$

$$n=2, b_2 = \frac{V_m}{2\pi} \left[1-1 + \cos \frac{10\pi}{3} - \cos \frac{4\pi}{3} \right] = 0$$

$$n=3, b_3 = \frac{V_m}{3\pi} \left[1+1 + \cos 5\pi - \cos 2\pi \right] = 0$$

$$b_n = \frac{200}{n\pi} \left[2 + \cos \frac{5\pi n}{3} - \cos \frac{2\pi n}{3} \right] ; n=1, 5, 7, 11, \dots$$

$$x(t) = 0 + \sum$$

$$x(t) = 0 + \frac{200}{\pi} \left\{ \sum_{n=1,5,7,11,\dots} \left\{ \frac{1}{n} \left[\sin \frac{2\pi n}{3} - \sin \frac{5\pi n}{3} \right] \cos n\omega_0 t + \left[2 + \cos \frac{5\pi n}{3} - \cos \frac{2\pi n}{3} \right] \sin n\omega_0 t \right\} \right\}$$

• 3rd order harmonics are absent

2.(a) Calculations for RMS values:

$$a_n = \frac{240\sqrt{2}}{\pi} \left[\frac{\cos(n+1)\alpha}{(n+1)} - \frac{\cos(n-1)\alpha}{(n-1)} + \frac{2(-1)^{n+1}}{n^2-1} \right]$$

$$b_n = \frac{240\sqrt{2}}{\pi} \left[\frac{\sin(n+1)\alpha}{(n+1)} - \frac{\sin(n-1)\alpha}{(n-1)} \right]$$

$$c_n = \sqrt{a_n^2 + b_n^2} \quad ; \quad \text{RMS} = \frac{c_n}{\sqrt{2}}$$

→ RMS of fundamental:

$$\begin{aligned} a_1 &= \frac{4}{2\pi} \int_{\alpha}^{\pi} 240\sqrt{2} \sin \omega t \cos \omega t \, d(\omega t) \\ &= \frac{2(240\sqrt{2})}{2\pi} \left[\int_{\alpha}^{\pi} \sin 2\omega t \, d(\omega t) + \int_{\alpha}^{\pi} \sin(0)\omega t \, d(\omega t) \right] \\ &= \frac{240\sqrt{2}}{2\pi} \left[-\cos 2\omega t \right]_{\alpha}^{\pi} = \frac{240\sqrt{2}}{2\pi} \left[-\cos 2\pi + \cos 2\alpha \right] \end{aligned}$$

$$a_1 = \frac{240\sqrt{2}}{2\pi} \left[-(-1)^2 + \cos 2\alpha \right] = \frac{240\sqrt{2}}{2\pi} \left[-1 + \cos 2\alpha \right]$$

$$\begin{aligned} b_1 &= \frac{240\sqrt{2}}{\pi} \int_{\alpha}^{\pi} 2 \sin \omega t \sin \omega t \, d(\omega t) \\ &= \frac{240\sqrt{2}}{\pi} \left[\int_{\alpha}^{\pi} \cos(0)\omega t \, d(\omega t) + \int_{\alpha}^{\pi} \cos 2\omega t \, d(\omega t) \right] \\ &= \frac{240\sqrt{2}}{\pi} \left[(\pi - \alpha) + \frac{\sin 2\omega t}{2} \right]_{\alpha}^{\pi} \\ &= \frac{240\sqrt{2}}{\pi} \left[(\pi - \alpha) + \frac{\sin 2\pi}{2} + \frac{\sin 2\alpha}{2} \right] \end{aligned}$$

$$b_1 = \frac{240\sqrt{2}}{\pi} \left[\pi + \frac{\sin 2\alpha}{2} - \alpha \right]$$

$$\text{Fundamental RMS} = \frac{\sqrt{a_1^2 + b_1^2}}{\sqrt{2}} = \frac{240}{\pi} \left[\left(\frac{-1 + \cos 2\alpha}{2} \right)^2 + \left(\pi - \alpha + \frac{\sin 2\alpha}{2} \right)^2 \right]^{\frac{1}{2}}$$

$\alpha = 30^\circ$

$$RMS = \frac{240}{\pi} \left[\left(\frac{-1}{2} + \frac{1}{4} \right)^2 + \left(\pi - \frac{\pi}{6} + \frac{\sqrt{3}}{4} \right)^2 \right]^{\frac{1}{2}} = \frac{238.8609}{\pi} V$$

$\alpha = 45^\circ$

$$RMS = \frac{240}{\pi} \left[\left(\frac{-1}{2} + 0 \right)^2 + \left(\pi - \frac{\pi}{4} + \frac{1}{2} \right)^2 \right]^{\frac{1}{2}} = \frac{221.5153}{\pi} V$$

$\alpha = 60^\circ$

$$RMS = \frac{240}{\pi} \left[\left(\frac{-1}{2} - \frac{1}{4} \right)^2 + \left(\pi - \frac{\pi}{3} + \frac{\sqrt{3}}{4} \right)^2 \right]^{\frac{1}{2}} = \frac{201.4016}{\pi} V$$

→ RMS of 3rd harmonic:

$$a_3 = \frac{240\sqrt{2}}{\pi} \left[\frac{\cos 4\alpha}{4} - \frac{\cos 2\alpha}{2} + \frac{1}{4} \right]$$

$$b_3 = \frac{240\sqrt{2}}{\pi} \left[\frac{\sin 4\alpha}{4} - \frac{\sin 2\alpha}{2} \right]$$

$$RMS = \frac{\sqrt{a_3^2 + b_3^2}}{\sqrt{2}} = \frac{240}{4\pi} \left[(\cos 4\alpha - 2\cos 2\alpha + 1)^2 + (\sin 4\alpha - 2\sin 2\alpha)^2 \right]^{\frac{1}{2}}$$

$\alpha = 30^\circ$

$$RMS = \frac{60}{\pi} \left[\left(\frac{-1}{2} - 2\left(\frac{1}{2}\right) + 1 \right)^2 + \left(\frac{\sqrt{3}}{2} - 2\left(\frac{\sqrt{3}}{2}\right) \right)^2 \right]^{\frac{1}{2}} = 19.0986$$

$\alpha = 45^\circ$

$$RMS = \frac{60}{\pi} \left[\left(-1 - 2(0) + 1 \right)^2 + \left(0 - 2(1) \right)^2 \right]^{\frac{1}{2}} = 38.1972$$

$\alpha = 60^\circ$

$$RMS = \frac{60}{\pi} \left[\left(\frac{-1}{2} - 2\left(\frac{-1}{2}\right) + 1 \right)^2 + \left(-\frac{\sqrt{3}}{2} - 2\left(\frac{\sqrt{3}}{2}\right) \right)^2 \right]^{\frac{1}{2}} = 57.2958$$